

CVLAB Summer Project

Stage 2: Flow Matching to Class Prototypes

Goal

In Stage 2, add a flow-matching (FM) layer to the prototype-based classifier from Stage 1. Given a frozen image feature z_i , the FM model should transport it toward the fixed prototype p_{y_i} of its class. Use the same datasets, encoders, class prototypes, training subsets, seeds, and test sets as in Stage 1. This is important so that the effect of the FM layer can be compared directly with the corresponding Stage 1 baseline.

The main goals are to:

- compare FM-based classification with the Stage 1 prototype baseline;
- compare standard FM training with rolled-out training;
- study how the learned transformation changes the geometry of the feature space.

Standard FM Training

For each training feature z_i and its class prototype p_{y_i} , sample $t \sim \mathcal{U}(0, 1)$, define $z_t = (1 - t)z_i + tp_{y_i}$, and use the target velocity $u_i = p_{y_i} - z_i$.

Train the velocity network $v_\theta(z, t)$ using

$$\mathcal{L}_{\text{FM}} = \|v_\theta(z_t, t) - u_i\|_2^2.$$

Thus, standard FM directly supervises the velocity prediction at randomly sampled points along the ideal interpolation path. At inference, however, the network is repeatedly evaluated on states generated by its own previous predictions.

Inference and Rolled-Out Training

At inference, start from the test feature $\hat{z}_0 = z$ and apply T Euler steps:

$$\hat{z}_{k+1} = \hat{z}_k + \frac{1}{T}v_\theta\left(\hat{z}_k, \frac{k}{T}\right), \quad k = 0, \dots, T-1.$$

Evaluate $T \in \{4, 12\}$. After the final step, classify $\hat{z}_T$ using the same class prototypes and cosine-similarity rule as in Stage 1.

In addition to standard FM training, implement a **rolled-out** version. Starting from each training feature $\hat{z}_0 = z_i$, apply the same full T -step sequence used at inference. The velocity network is unchanged: at every step it still predicts $v_\theta(\hat{z}_k, k/T)$.

The difference is in the training objective. Rolled-out training exposes the network to the same sequence of self-generated intermediate states that appears at inference. Instead of supervising the individual velocity predictions, optimize how close the final transported feature is to the target prototype:

$$\mathcal{L}_{\text{roll}} = \|\hat{z}_T - p_{y_i}\|_2^2.$$

Backpropagate through the complete sequence of T velocity predictions. For rolled-out training, use the same value of T during training and inference.

Keep the network architecture and the main training choices fixed when comparing standard FM and rolled-out training.

Velocity Network

Use a relatively small MLP for $v_\theta(z, t)$. A reasonable suggestion is 2 hidden layers of width around 512, SiLU activations, scalar time t concatenated to the input feature, and output dimension equal to the feature dimension.

There is no need for extensive architecture or hyperparameter optimization. The goal is to obtain stable training and a fair comparison between the Stage 1 baseline, standard FM, and rolled-out training.

Experimental Protocol

Use the same training-set sizes as in Stage 1, $K \in \{5, 10, \text{full}\}$, with the same sampled subsets and seeds. For each setting, compare:

- Stage 1 prototype baseline;
- standard FM with $T = 4$ and $T = 12$;
- rolled-out FM with $T = 4$ and $T = 12$.

Use top-1 accuracy on the complete official test split and follow the same repetition protocol as in Stage 1.

Results to Present

You should be prepared to present and discuss:

1. **Classification results.** Compare the Stage 1 baseline with standard and rolled-out FM for both values of T and for $K = 5, 10, \text{full}$. Report top-1 accuracy and also indicate the change relative to the corresponding baseline, $\Delta\text{Acc} = \text{Acc}_{\text{FM}} - \text{Acc}_{\text{baseline}}$. A table and/or an accuracy-versus- K plot with error bars should make the comparison clear.
2. **Training curves.** Show representative training-loss curves for standard FM and rolled-out training. The main purpose is to verify that training is stable and that both approaches reach reasonable solutions.
3. **Feature-space visualizations.** For a readable subset of classes, compare the original encoder features, the features after standard FM, and the features after rolled-out FM. Use the same test examples, class colors, and corresponding class prototypes across the compared plots. PCA or t-SNE may be used. Compute the projection jointly over the feature sets and prototypes being compared so that the different views correspond to the same low-dimensional representation.
4. **Flow trajectories.** For a small number of representative test examples, visualize several intermediate FM states together with the original feature, final transported feature, and corresponding class prototype. PCA is recommended for this visualization because the projected trajectories are easier to interpret geometrically.

Optional. You are also encouraged to explore the learned flow in the reverse direction, starting from the class prototypes, and to consider comparing samples and prototypes at intermediate flow times.